

Name: Sagi Sahasra Srihasa
Roll : 2520030008

Step 1:

```
1  /*
2      Name: Sagi Sahasra Srihasa
3      Roll : 2520030008
4  */
5  ● CREATE DATABASE BankDB;
6  ● USE BankDB;
7
8  ● CREATE TABLE Customer (
9      Customer_ID INT PRIMARY KEY,
10     Customer_Name VARCHAR(100) NOT NULL,
11     Phone VARCHAR(15),
12     Email VARCHAR(100),
13     City VARCHAR(50)
14 );
```

#	Time	Action
✓ 1	11:07:44	CREATE DATABASE BankDB
✓ 2	11:07:46	USE BankDB
✓ 3	11:07:49	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU

Step 2:

```
● CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2) DEFAULT 0,
    Branch VARCHAR(50),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);

● CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (Account_No) REFERENCES Account(Account_No)
);
```

✓	4	11:10:30	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20) NOT NULL, Amount INT NOT NULL);	0 rows affected
✓	5	11:10:33	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction_Type VARCHAR(20) NOT NULL, Amount INT NOT NULL);	0 rows affected

Step 3:

```
INSERT INTO Customer
VALUES
(101, 'Sahasra', '9876543210', 'sahasra@gmail.com', 'Hyderabad'),
(102, 'Priya ', '9876543211', 'priya@gmail.com', 'Vijayawada'),
(103, 'Karan Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
(104, 'Nehitha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
(105, 'Kiran ', '9876543214', 'kiran@gmail.com', 'Hyderabad');
```

```
INSERT INTO Account
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');
```

✓	6	11:11:06	INSERT INTO Customer VALUES (101, 'Sahasra', '9876543210', 'sahasra@gmail.com', 'Hyderabad'), (102, 'Priya ', '9876543211', 'priya@gmail.com', 'Vijayawada'), (103, 'Karan Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'), (104, 'Nehitha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'), (105, 'Kiran ', '9876543214', 'kiran@gmail.com', 'Hyderabad');	5 rows affected
✓	7	11:11:15	INSERT INTO Account VALUES (10001, 101, 'Savings', 50000, 'Hyderabad'), (10002, 102, 'Savings', 75000, 'Vijayawada'), (10003, 103, 'Current', 120000, 'Bangalore'), (10004, 104, 'Savings', 45000, 'Chennai'), (10005, 105, 'Current', 90000, 'Hyderabad');	5 rows affected

Step 4:

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000);
```

✓	8	11:11:39	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOSIT', 15000), (10003, 'WITHDRAW', 20000);	3 rows affected
---	---	----------	--	-----------------

Step 5:

```
DELIMITER //
```

```
CREATE PROCEDURE GetAllCustomers()
```

```
BEGIN
```

```
    SELECT * FROM Customer;
```

```
END //
```

```
CREATE PROCEDURE GetAccountDetails(
```

```
    IN p_Account_No INT
```

```
)
```

```
BEGIN
```

```
    SELECT *
```

```
    FROM Account
```

```
    WHERE Account_No = p_Account_No;
```

```
END //
```

✓	9	11:12:49	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	0 row(s) affected
✓	10	11:12:58	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account ...	0 row(s) affected

Step 6:

```
CREATE PROCEDURE DepositMoney(
```

```
    IN p_Account_No INT,
```

```
    IN p_Amount DECIMAL(12,2)
```

```
)
```

```
BEGIN
```

```
    IF p_Amount <= 0 THEN
```

```
        SIGNAL SQLSTATE '45000'
```

```
        SET MESSAGE_TEXT = 'Amount must be greater than zero';
```

```
    ELSE
```

```
        UPDATE Account
```

```
        SET Balance = Balance + p_Amount
```

```
        WHERE Account_No = p_Account_No;
```

```
    END IF;
```

```
END //
```

✓	11	11:13:33	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN I...	0 row(s) affected
---	----	----------	--	-------------------

Step 7:

```
CREATE TRIGGER CheckBalance
BEFORE UPDATE ON Account
FOR EACH ROW
BEGIN
    IF NEW.Balance < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Insufficient balance';
    END IF;
END //
```



```
CREATE TRIGGER CheckTransactionAmount
BEFORE INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    IF NEW.Amount <= 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';
    END IF;
END //
```


✓	14	11:14:24	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Bala.
✓	15	11:14:29	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE.

Step 8:

```
CREATE TRIGGER UpdateBalanceAfterTransaction
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    IF NEW.Transaction_Type = 'DEPOSIT' THEN
        UPDATE Account
        SET Balance = Balance + NEW.Amount
        WHERE Account_No = NEW.Account_No;
    ELSEIF NEW.Transaction_Type = 'WITHDRAW' THEN
        UPDATE Account
        SET Balance = Balance - NEW.Amount
        WHERE Account_No = NEW.Account_No;
    END IF;
END //
```


✓	16	11:16:07	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH ROW... 0 row(s) affe
---	----	----------	---

Step 9:

```
125 • CALL GetAllCustomers();
```

```
126
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Sahasra	9876543210	sahasra@gmail.com	Hyderabad
	102	Priya	9876543211	priya@gmail.com	Vijayawada
	103	Karan Reddy	9876543212	arjun@gmail.com	Bangalore
	104	Nehitha Rao	9876543213	sneha@gmail.com	Chennai
	105	Kiran	9876543214	kiran@gmail.com	Hyderabad

Step 10:

```
127 • CALL GetAccountDetails(10001);
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Account_No	Customer_ID	Account_Type	Balance	Branch
	10001	101	Savings	50000.00	Hyderabad

Step 11:

```
CALL DepositMoney(10001, 5000);
```

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2500);
```

✓	19	11:17:30	CALL DepositMoney(10001, 5000)	1
✓	20	11:17:45	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 25...	1

Step 12;

137

138 • `SELECT * FROM Account;`

Result Grid  Filter Rows: Edit:    Export/Import:   Wrap Cell Content					
	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	57500.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
Account 3					

Step 13;

140 • `SELECT * FROM Bank_Transaction;`

Result Grid  Filter Rows: Edit:    Export/Import:   Wrap Cell Content					
	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
▶	1	10001	DEPOSIT	10000.00	2026-08-15 11:11:39
	2	10002	DEPOSIT	15000.00	2026-08-15 11:11:39
	3	10003	WITHDRAW	20000.00	2026-08-15 11:11:39
	4	10001	DEPOSIT	2500.00	2026-08-15 11:17:45
*	NULL	NULL	NULL	NULL	NULL
Bank Transaction 4					